

// BYE-BYE LOCK-IN: AUTOMATE CLOUD SWITCHING USING OPEN SOURCE

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But Why?

Tool Stacks



PM+Docs



Software Dev

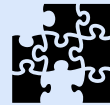
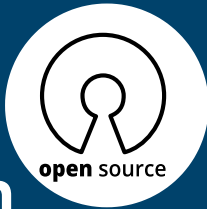


DevOps

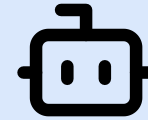
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EcoSystem



Integrations



AI Assistant



Low Ops

Infra



K8s

But Why?

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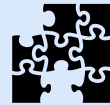
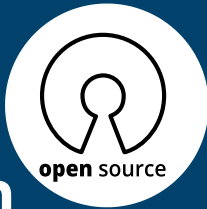


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DevOps

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VM



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Automatic Migration



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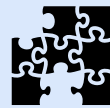
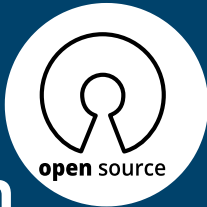
Software Dev



DevOps

...

EcoSystem



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K8s



Cloud
Providers



On
Prem



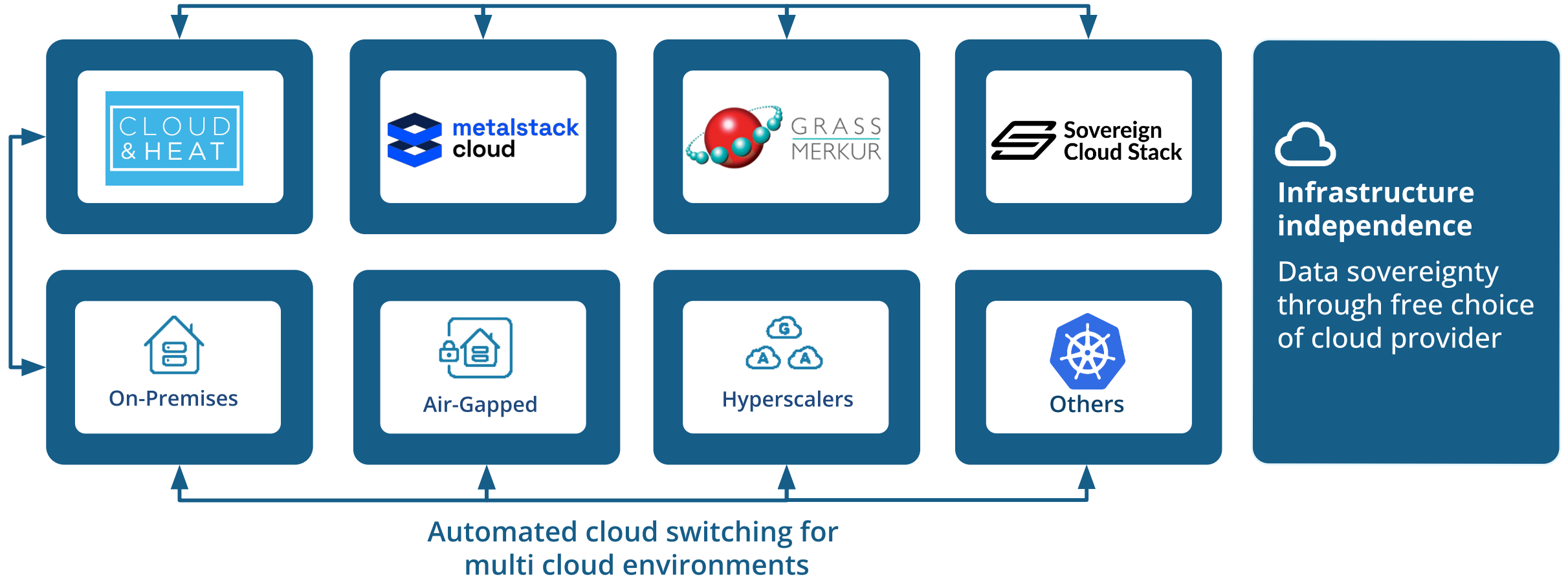
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Automatic Migration



Automatic Cloud Switching

Infra independence through cloud switching



Reasons to switch

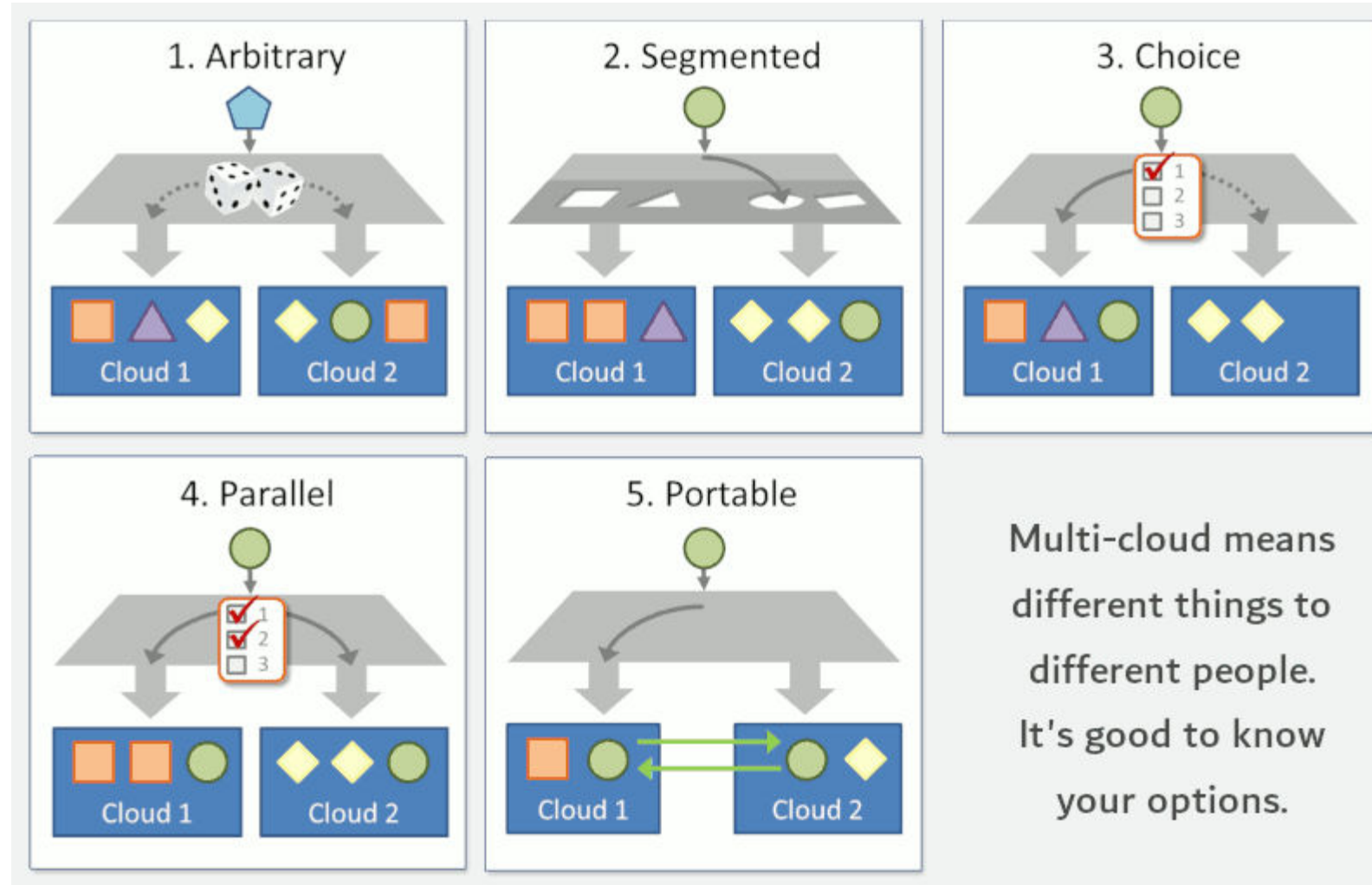
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- ☁ ↔ 🏠
- ...

!! Ability to switch (digital sovereignty)

➡ Multi Cloud



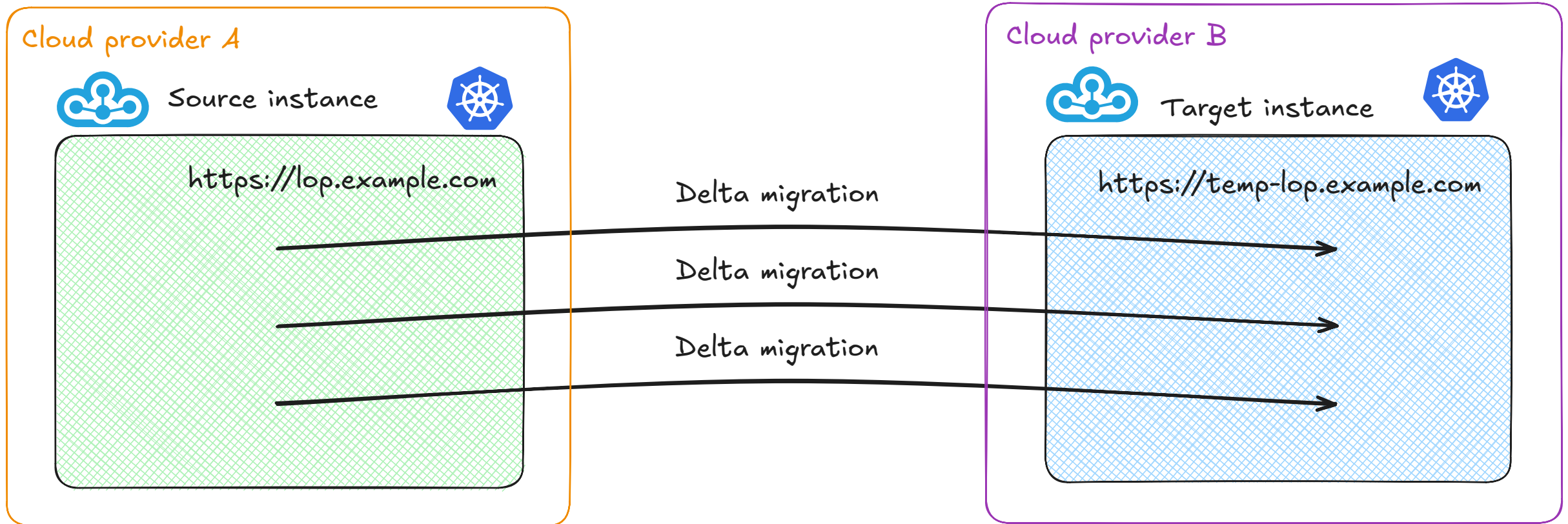
Multi Cloud Architecture Options



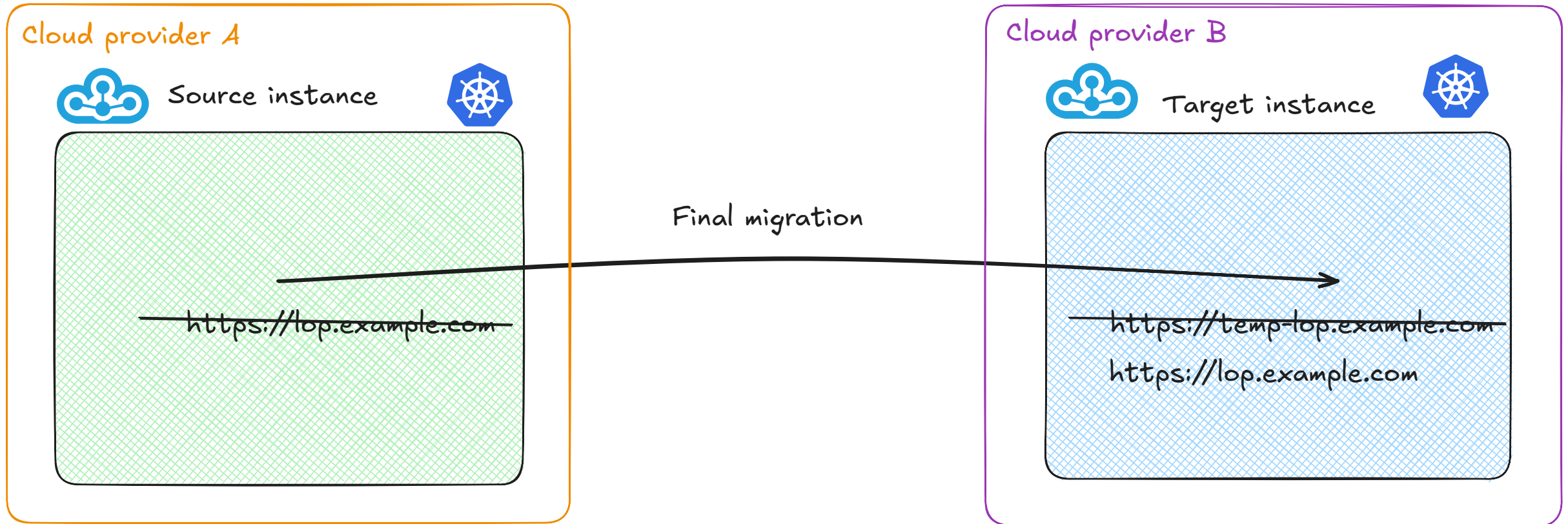
Automatic Cloud Switching



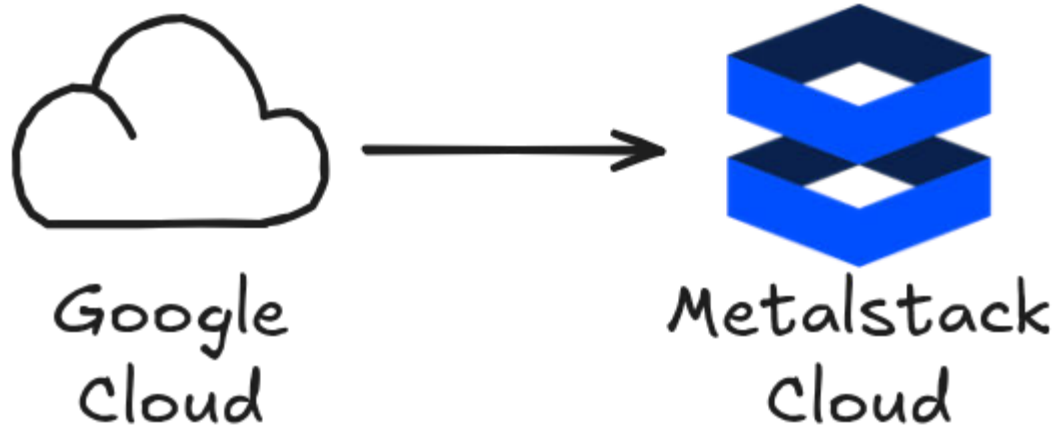
Delta migration



Final migration



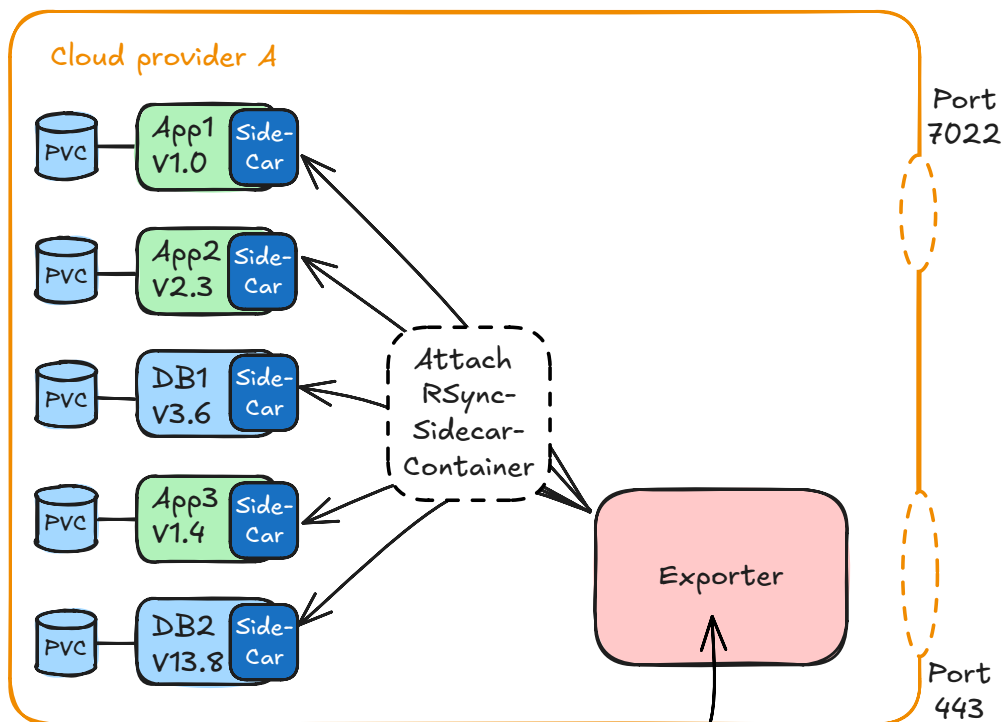
Demo migration



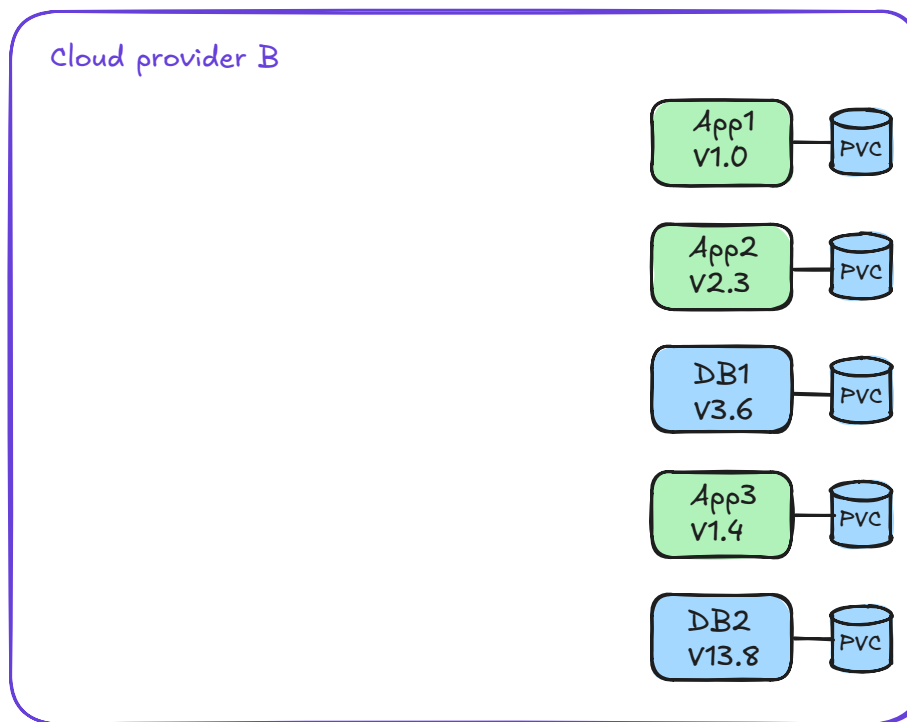
Preparing the migration



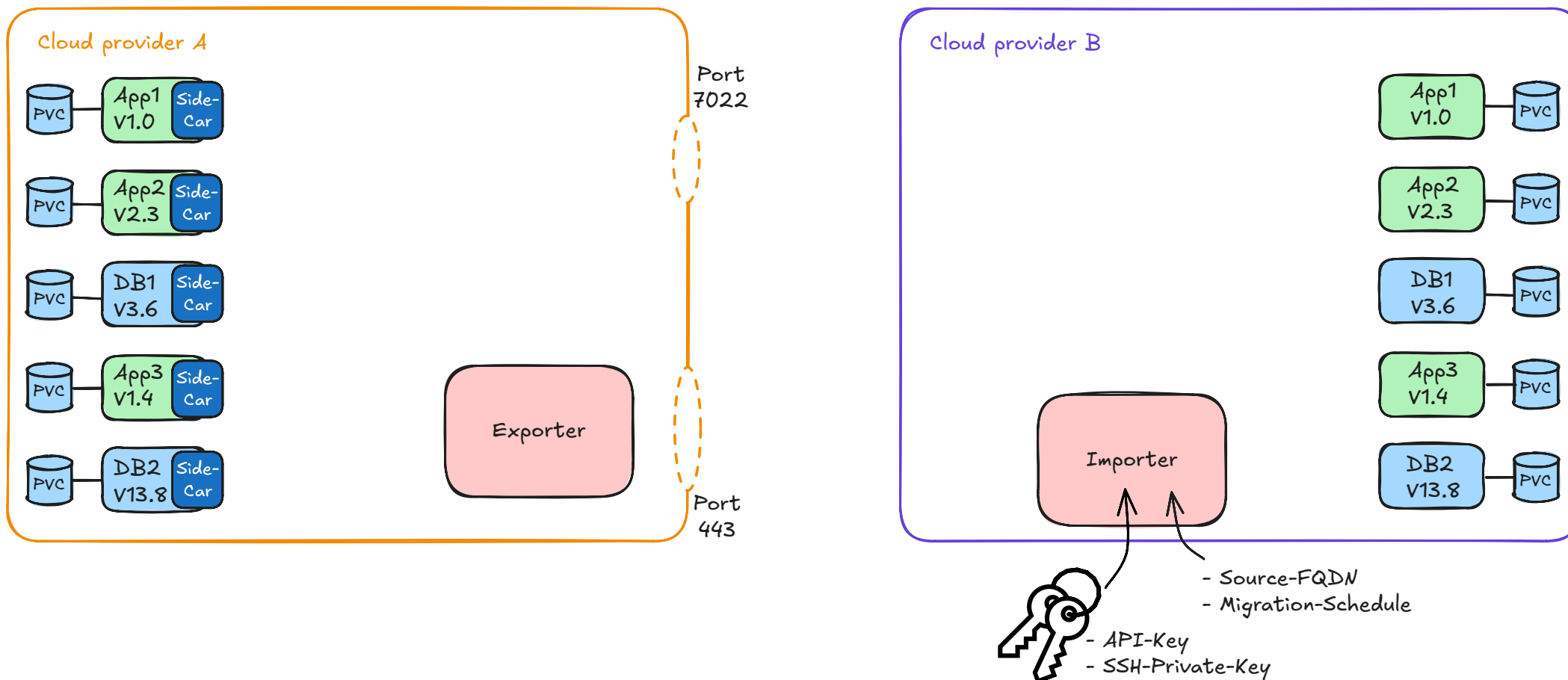
Prepare source instance



- API-Key
- SSH-Public-Key



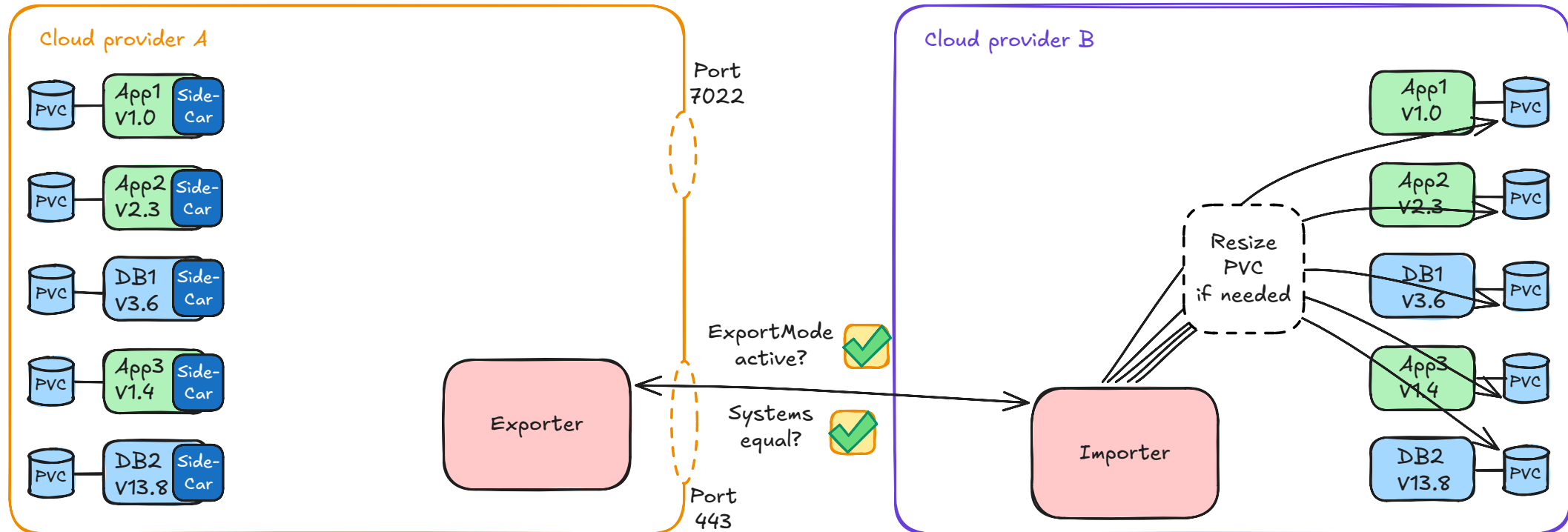
Prepare target instance



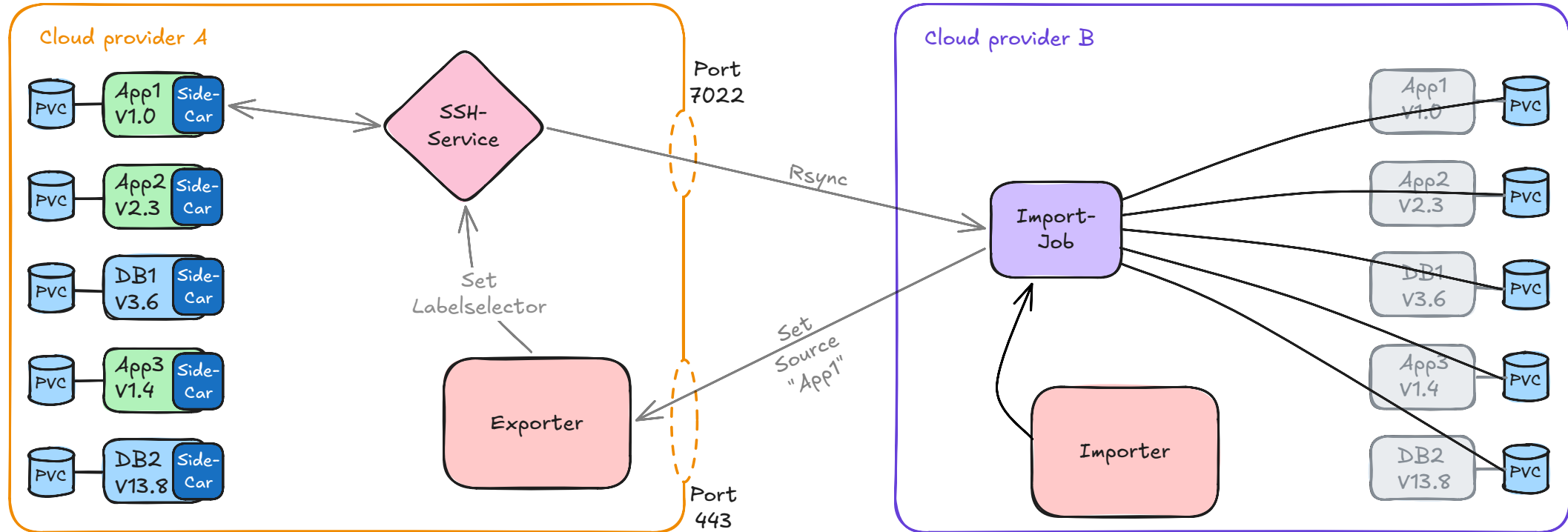
Executing the migration



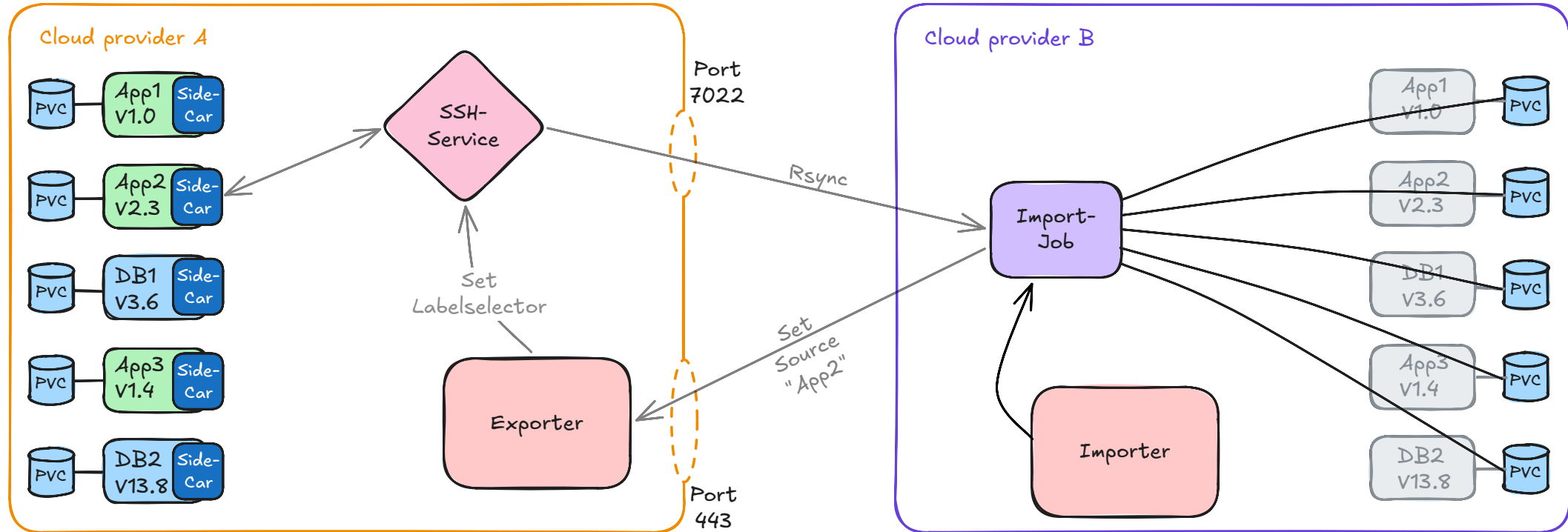
Verify preconditions



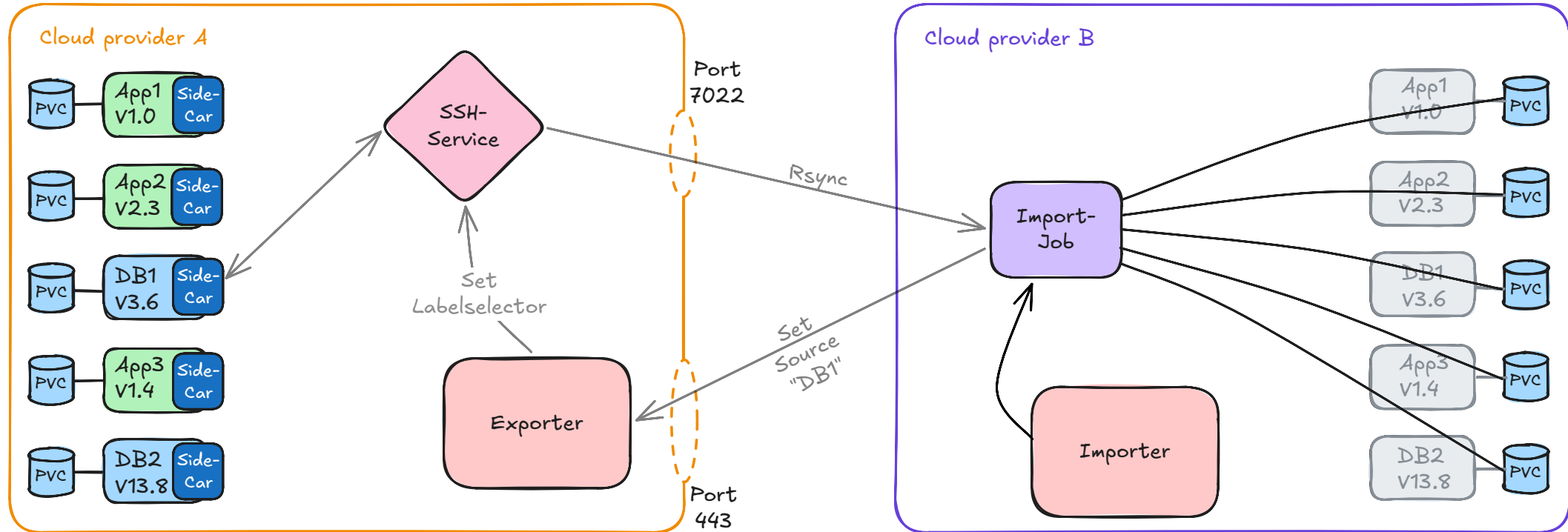
Migrate volume data



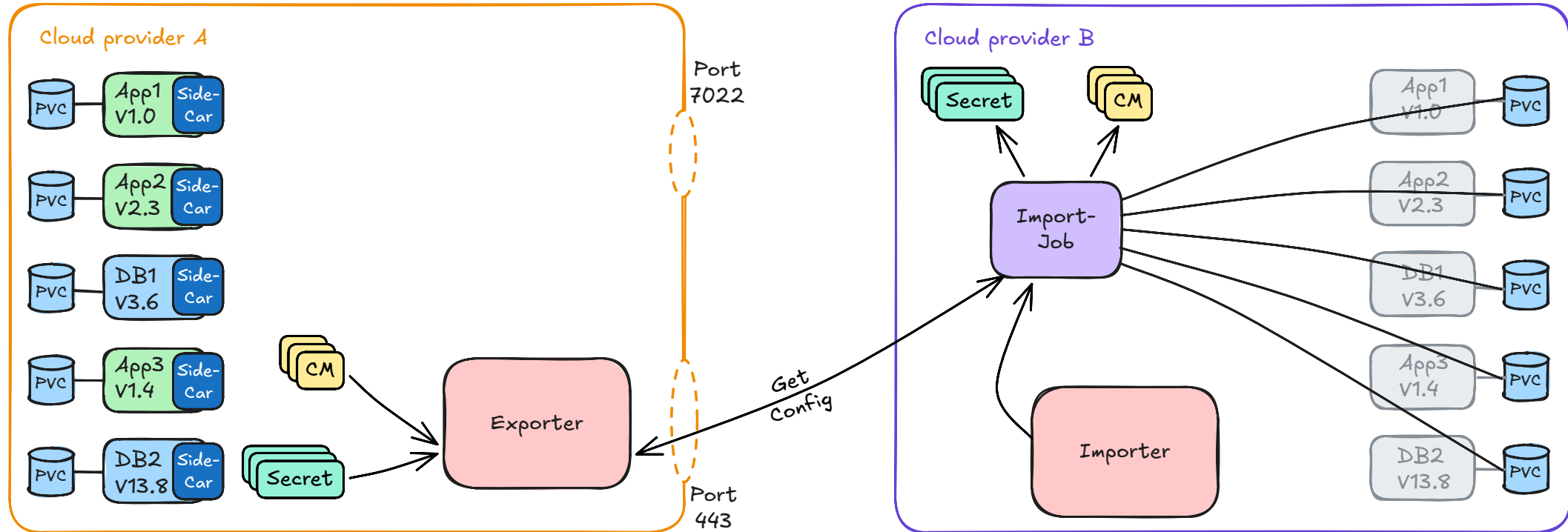
Migrate volume data



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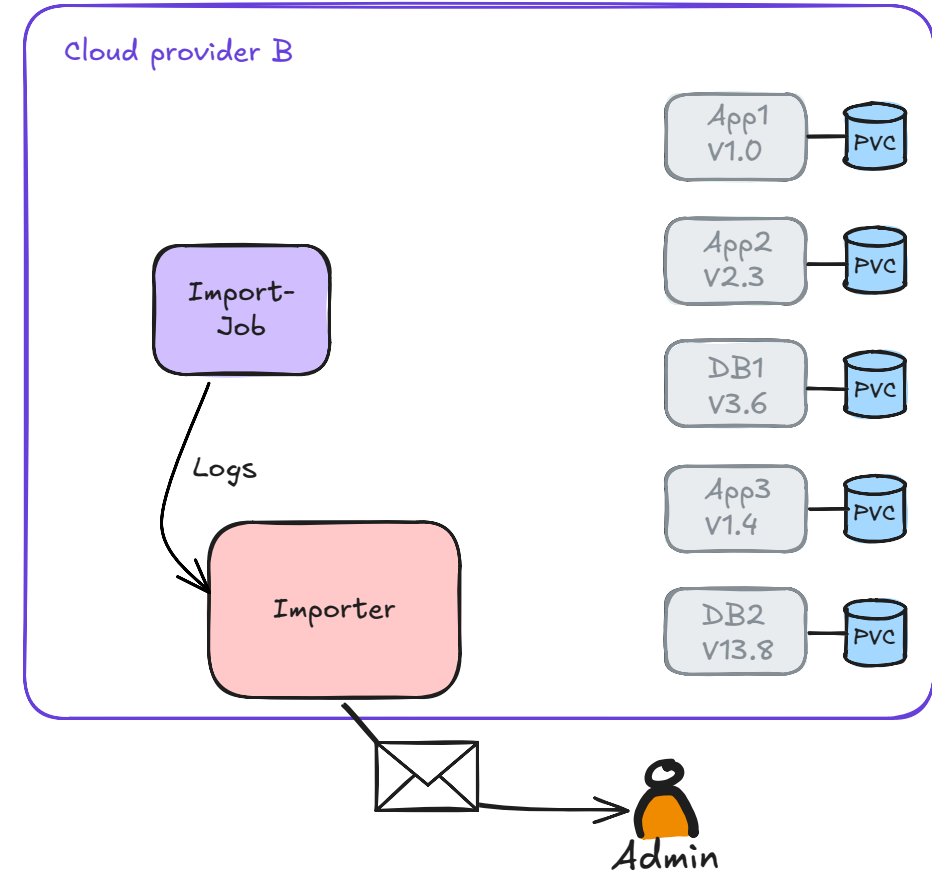
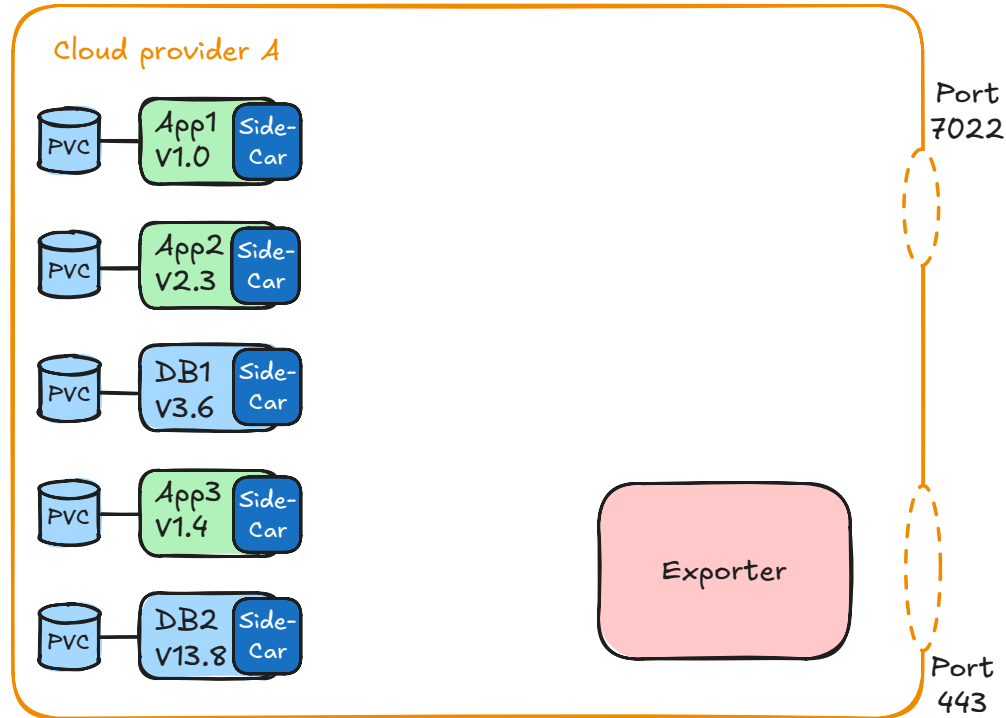
Migrate configuration data



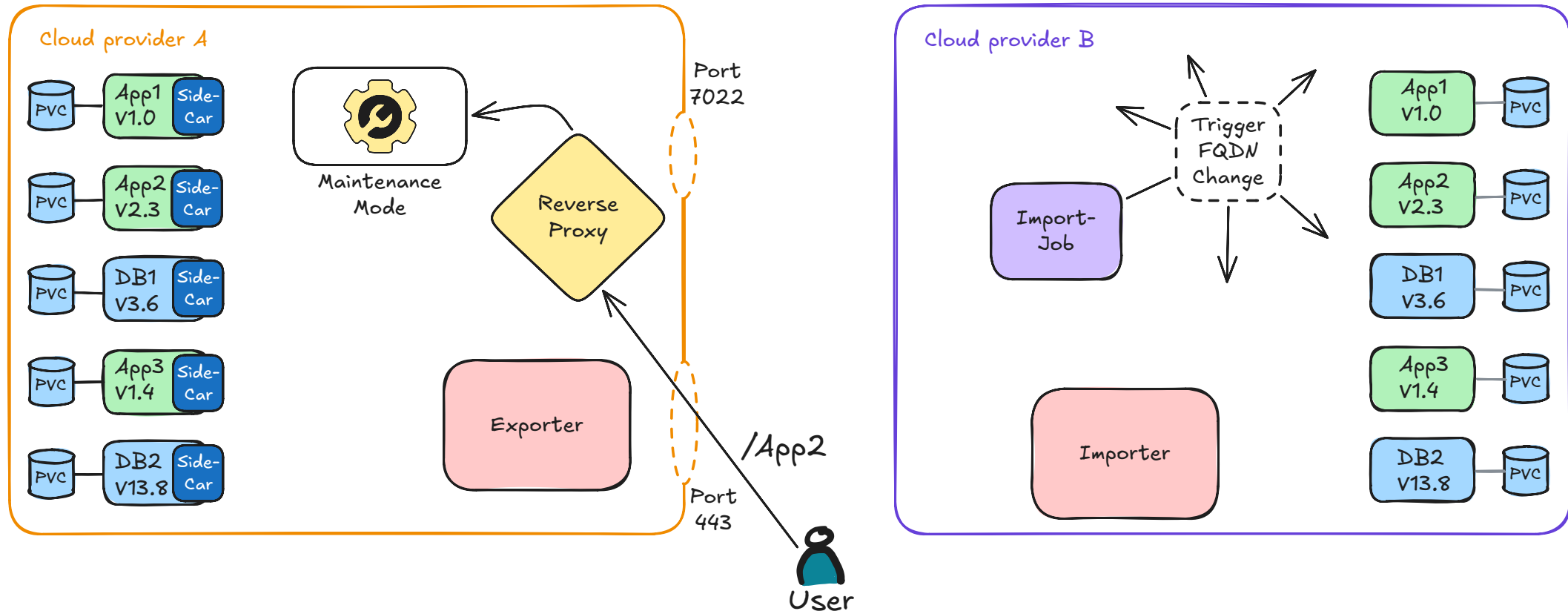
Finishing the migration



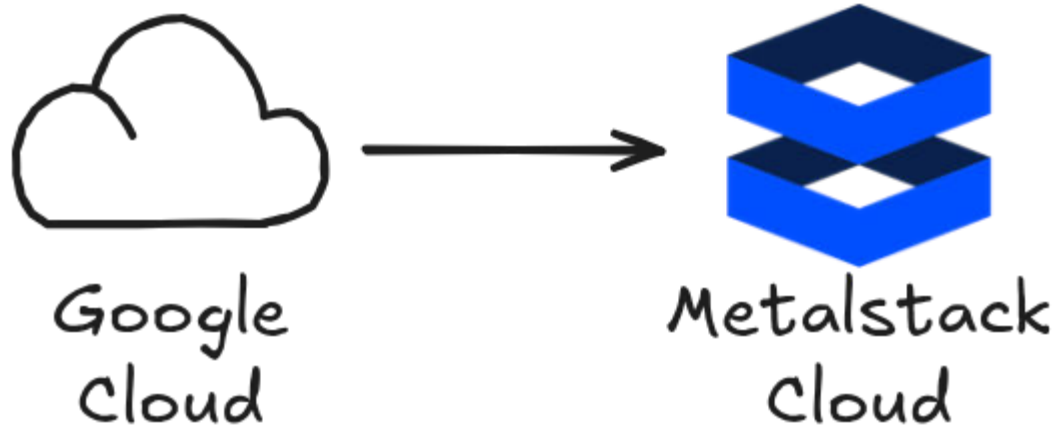
Finish migration



Additional steps on final migration



Demo migration



Conclusion



Migration without the usual pain

-  Repeatable migrations → safe to test
-  Delta-based transfer with rsync
-  Minimal network requirements (443 + SSH)

What we learned

-  PoCs helped de-risk key decisions
-  Separation (exporter / importer) kept it flexible
-  Manual testing became a bottleneck

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 cloudogu/ces-exporter  cloudogu/ces-importer

Image sources

- Automatic Cloud Switching
https://www.freepik.com/free-vector/cloud-network-technology-icon-white-gradient-background_16406372.htm
- Buzzwords
<https://makeameme.org/meme/buzzwords-buzzwords-everywhere-a120508e67>